

Pressure- and solenoid-operated air control valves (Figure 19) are used in general and specialist services including steam, combustion gases, cryogenics, vacuum, dust collector systems, engineering, proportional, and explosive atmospheres, etc.

Figure 20 shows how pneumatic-distribution spool-valve islands can be connected with a PLC or PC control system through a multiwire cable or with a fieldbus through a communications protocol. These systems meet the needs for automated installations and allow the transmission of any control signal to the spool valves and any information signal from the position detections. A typical connection structure is shown in Figure 21.

See also the chapter on Actuators.

Electromagnetic control valves

Typically, electromagnetic control valves are solenoid-actuated and employ a pilot disc to assist operation. They do not use any stem, packing or bellows and there are few moving parts.



Figure 17. Iris-type diaphragm control valve.

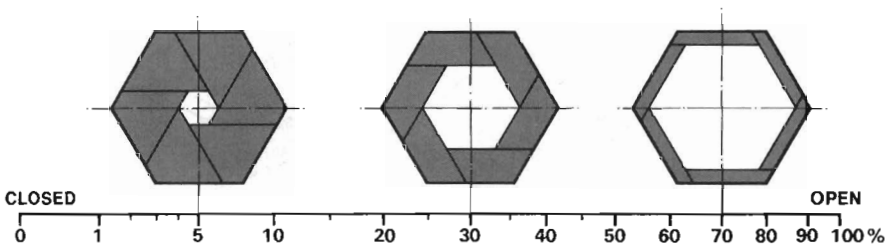


Figure 18. Iris-type diaphragm control valve: opening sequence.

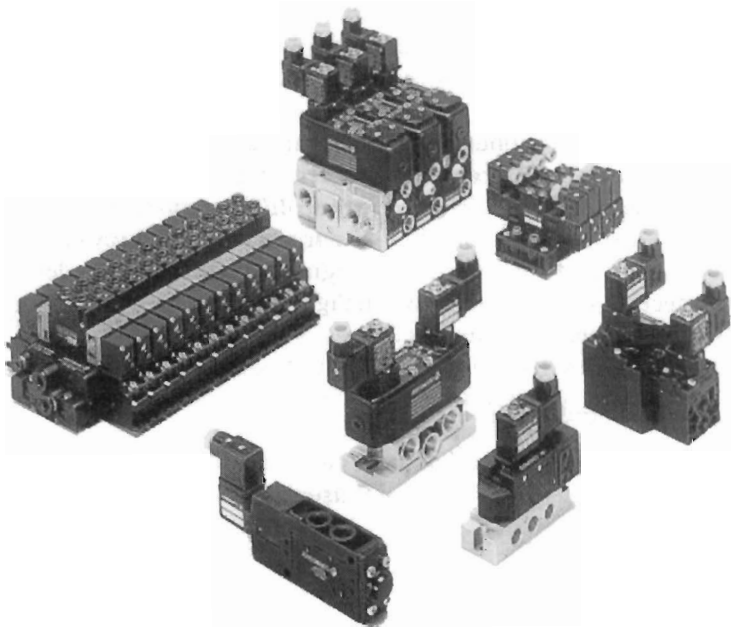


Figure 19. Solenoid air poppet and spool valves.

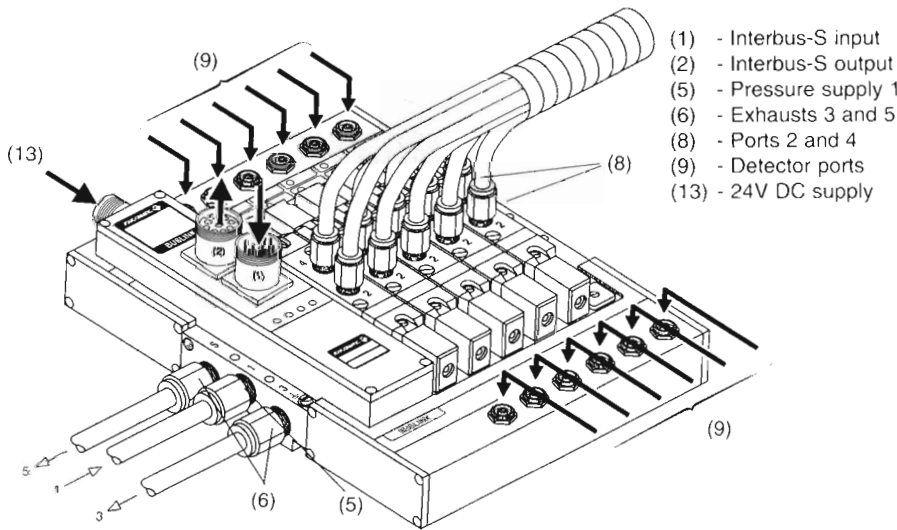


Figure 20. Spool-valve island with fieldbus connection to PLC or PC control system.

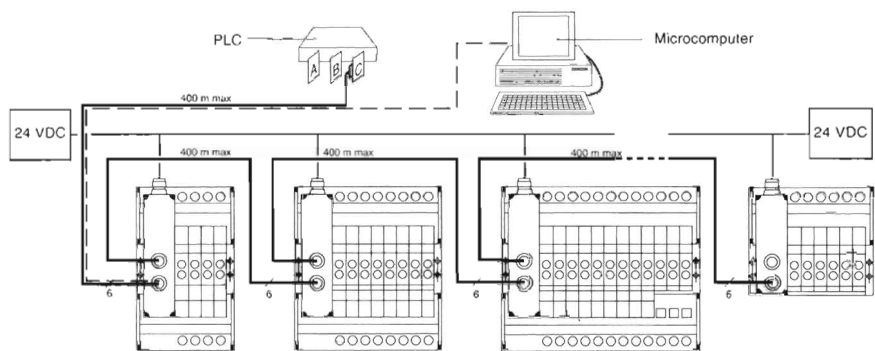


Figure 21. Typical Interbus-S connection structure.

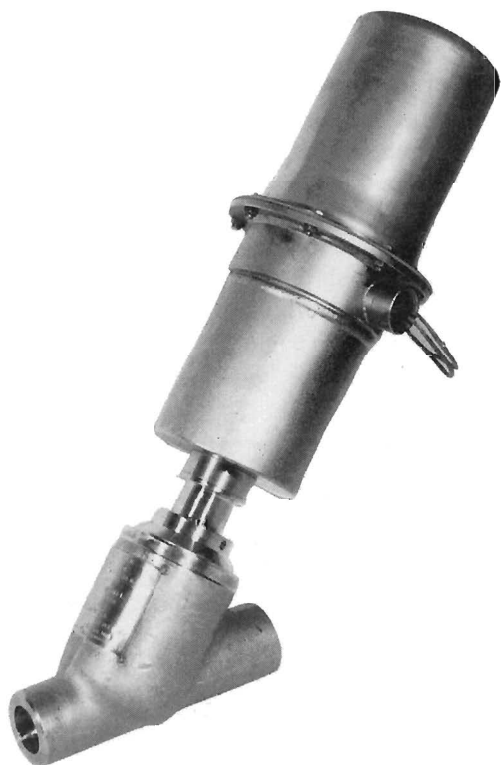


Figure 22. Electromagnetic control valve—'equivalent to pipe'.

Two basic designs are on-off isolation valves and modulating control valves. Operation of the valve occurs when the solenoid is energised to develop a magnetic field. This lifts the plunger and pulls a pilot off its seat to open an internal vent port. When the differential pressure between the top and bottom surfaces of the main disc changes sufficiently, the main disc follows in servo motion to the pilot and allows the fluid at the inlet to flow.

In modulatory designs, an electronic positioner controls electrical power to the solenoid to match disc position with desired flow.

An electromagnetic control valve is shown in Figure 22. Typical applications include storage and handling of hydrocarbons.

The increasing use of rotary valves proceeds hand in hand with the widening application range, improved reliability and cost-effectiveness of control valves. It is also becoming standard practice to combine globe valves in small sizes with rotary valves in larger sizes. In on-off applications, the rotary types are used throughout the size range. The availability of manufacturers' software programs for sizing control valves provides a valuable tool for the engineer who needs to evaluate the true performance of the control valve. It can also give information on how to program the controller to achieve the required control characteristic and thus to improve control accuracy and overall process-cost efficiency.

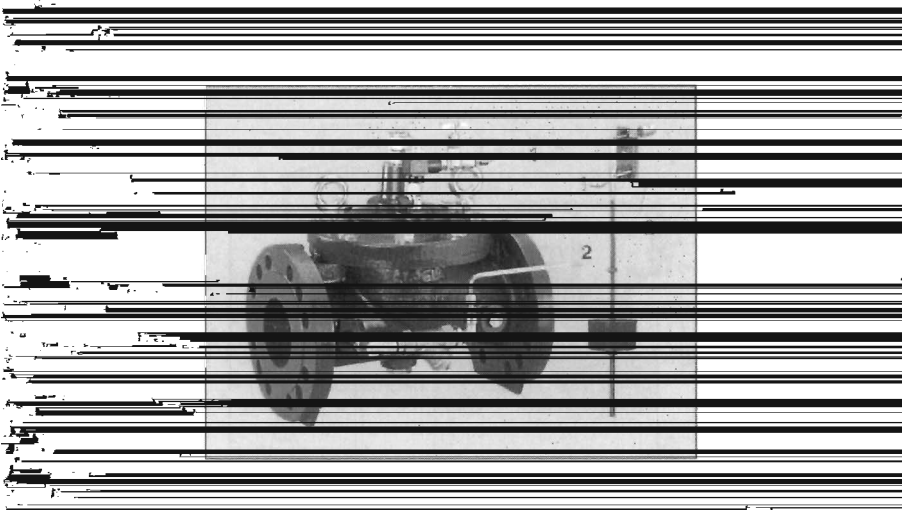
Float Control Valves

The traditional type of float control valve is the simple on–off valve controlled by a float: see the chapter on Ball Float Valves (Section 2). The float principle is also widely used for other forms of level control valves, either directly or indirectly.

With indirect operation, the float movement may be used to operate a pilot valve, which in turn operates a remote main control valve, or merely to generate air pressure on a piston-cylinder basis proportional to the rise in liquid level. At a predetermined point, this pressure becomes sufficient to operate the level control valve.

When line pressure is used to operate the valve, tubing connections are made from the float control pilot to the valve cover, and also to the inlet side of the valve. The control may be installed at any elevation provided the flowing line pressure in bar (lbf/in^2) is greater than the head in metres (feet) between the valve and valve control.

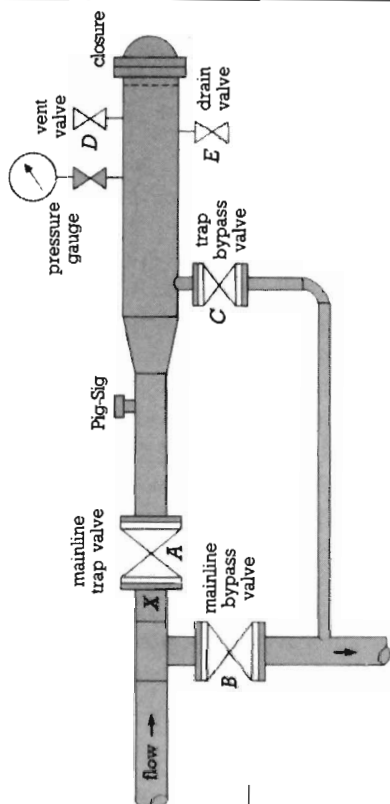
An independent source of either air or water can also be used to operate the valve. In this case, the pressure from this independent source must be equal to, or greater than, the pressure at the valve inlet. The independent source is connected to the float control pilot in place of the line running from the inlet



receiving

Starting Condition:

Trap is empty at atmospheric pressure. Valves B, D and E are open. Valves A and C are closed. Closure door is closed and secured.



1. To purge the trap, close Valve E and slowly open Valve C.
2. After purging, allow the trap pressure to equalize by closing Valve D with Valve C open.
3. With Valve C still open, open Valve A. The trap is now ready to receive the pig.
4. When the pig arrives, it may stop between trap Valve A and the tee (point X).
5. Partially close Valve B. This will force the pig into the trap due to increasing flow through Valve C.
6. After the pig is in the trap as indicated by the Pig-Sig, open Valve B and close Valves A and C.
7. Open Valves D and E and vent the trap to atmospheric pressure.
8. After the trap is vented (0 psig) and drained with Valves D and E open, open the closure door and remove the pig.
9. Close and secure the closure door.

Figure 4. —continued.

Modern pigging systems now operate with a 'captive pig' and the pipeline is opened up only occasionally to check on the condition of the pig. At all other times, the pig is shuttled up and down the pipeline at the end of each transfer, for example.

Today's pigging systems can also be operated by a Programmable Logic Controller (PLC) or other computer-based control system.

System types

Pigging systems usually comprise one or more of the following types:

1. Open system: where the pigs are forced into and expelled from the system whenever necessary. The system works on a one-way principle.
2. Closed system: the pigs remain 'captive' in the system for their entire operational life and do not need to be forced into and expelled from the pipe.
3. Single-pig system: has only one pig in the pipe which goes into action when, for example, product residue needs to be cleared out of the line. This can be a cost-effective answer for simple tasks.
4. Two-pig system: probably the pigging system chosen most often. It has two pigs in a pig launching and receiving station. When the pumping process begins, the first pig is pushed into the line and the product then pumped in after it. The second pig pushes the product out of the pipe, cleaning any residue from the pipe as it moves along.

Smart pigs

Smart pigs are generally deemed to be any series of pigs used for internal inspection of a pipeline. Smart pigs can detect bends, dents and other reductions, inspect for corrosion and other damage and photograph the internal walls of a pipeline. Sophisticated electronics and computers in sealed containers take and record various internal measurements.

Enhanced cleaning pigs

These pigs are specially designed to be more aggressive than standard cleaning pigs.

Typically, there are three types:

- Magnetic cleaning pig for dealing with loose debris in both gas and liquid lines.
- Pin wheel-type pig for removing hard scale and wax deposits adhering to the pipe wall, usually in liquid lines.

- Brush wheel pig which is run after a line has been enhanced and cleaned using a magnetic cleaning pig, a pin-wheel pig, or a combination of both.

Enhanced cleaning pigs can be set up for various levels of aggressiveness, ranging from very low to very high. This gives the advantage of implementing controlled and gradual increases in aggressiveness to prevent over-cleaning, which could lead to a blocked pipeline.

Foam pigs are made of polyether or polyurethane foam from 32 to 144 kg/m³ (2 to 9 lbf/ft³) density, chamfered on the leading edge and with a sealing disc on the rear pressure face. On some types the outside surface is coated with elastomeric polyurethane to increase the resistance to abrasion from the pipe wall. Abrasive foam pigs, in addition, have bands of abrasive grit around the circumference to improve the cleaning action. While normally made with a circular cross-section, foam pigs can be fabricated with other profiles for cleaning ductwork.

Steel pigs are, in comparison, relatively complex, consisting of a steel frame which seals against the pipe wall by means of two or more elastomeric cup seals. Except in the case of gauging and bi-directional pigs, cleaning is achieved by a series of scrapers or brushes arranged around the outside of the frame. Gauging pigs consist simply of the frame and seals, while the seals of bi-directional steel pigs are placed in opposition to each other to enable the pig to be driven in either direction.

Table 4 summarises the principal applications of each type and Table 5 gives recommendations for certain pigs in a given application.

Table 4.

Type of pig	Principal uses	Example
Foam	Swabbing; removal of thin, soft or liquid films	Sludge or water in air lines; to avoid damaging pipe surfaces.
Coated foam	Removal of films and heavy sludges	Oily deposits.
Abrasive foam	Removal of heavy or hard deposits	Rust; slag; hard scale in water; effluent and hydrocarbon lines.
Cleaning (brush)	Removal of miscellaneous superficial materials	Light rust and scale; pre-commission cleaning.
Scraper	Removal of hard or strongly adhered films and deposits	Corrosion; wax in effluent and hydrocarbon lines.
Gauging	Proving minimum bore prior to commissioning	
Bi-directional	Plugging pipeline during hydrostatic testing	

Table 5. Pigging application

Pipeline application	Pig choices
Geometry survey, e.g. proving that the inside diameter of the line has a minimum clear bore	Gauging or caliper pig.
Removal of rust, scale, and hard deposits	Brush cleaning pig.
Removal of wax-type deposits	Urethane-bladed cleaning pig.
Chemical swabbing	Spheres; multi-cupped pig.
Water filling/air removal	Bi-directional pig; foam pig; multi-cupped batching pig.
Dewatering Higher cost: maximum dewatering Lower cost: less dewatering	Cup swabbing, bi-directional foam pigs.
Product removal	Multi-cupped pig; foam pig; spheres.
Internally-coated lines	Foam pig; urethane brush pig.
Product separation	Spheres; multi-cupped batching pig.
Meter proving	Spheres.
Corrosion/material defect	Corrosion-detection pig.
On-stream liquid removal	Spheres; multi-cupped pigs.

Valves

Full-bore, full-opening valves in accordance with API-6D standards are required for safe pig passage. Reduced port valves usually cannot be pigged.

Some common piggable valves include: through-conduit gate valves, wedge-type gate valves, ball valves and check valves.

Wedge gate valves can sometimes present problems for pigs, e.g. those that are gapped between seat ring spacings. Generally, ball valves are the best type for use in pipelines that will be pigged.

Optimum cleaning methods

Careful selection of equipment is essential if a pigging operation is to be successful, and this depends upon possessing adequate information on the following factors:

- internal pipe diameter
- bend radii

- location and type of branches, valves, flow meters and other obstructions
- location of access points
- material of construction
- pressure rating of pipelines
- constraints on propellant type
- physical properties of material to be removed
- distribution and thickness of material to be removed
- health hazards

Nominal pipe bore, actual bore and bend radii are of crucial significance in selecting the sizes and types of pigs to be used. Very tight bends or changes in direction can be negotiated only by foam pigs. Where there is a possibility of the pig becoming stuck, it is necessary to use a bi-directional pig so that it can be withdrawn from the blockage.

The physical properties and distributions of the material to be removed influence the type of cleaning action required and hence the choice of pig. In many cases, a range of different pigs will be used in succession to provide the optimum cleaning action at each stage of the operation. Unfortunately, it is often difficult to determine in advance which pigs will be needed, and it is sometimes necessary to make the choice of pig types as the operation proceeds.

It is essential to identify beforehand all intersections of pipe runs, change of bore and obstructions in the bore from valves, pumps and monitoring equipment, and to plan the pigging operations carefully.

In complex process pipework it may be necessary to break the operation into separation sections. On long pipelines there is no reason, however, why a pig should not be sent many kilometres (miles). If desired it can be followed electrically or magnetically by the use of tracking equipment.

Applications

Pigging as a technique is suitable for cleaning the walls of pipes and removing sediment from the invert. It is particularly suitable for long runs of pipework having many bends and few access points, which cannot easily be cleaned by

Table 6.

Pig type	Range of actual bore		Minimum bend radius	Uni/bi-directional
	D (mm)	D (in)		
Foam	12–1220	$1/2$ –48	No limits	Bi-directional
Steel	50–1420	2–56	$1\frac{1}{2}D$ –5D	Either

any other method. Pigging is not suitable for unblocking pipes which are completely closed.

With the correct choice of pigs and equipment, pigging can be successful in dealing with a wide variety of materials of which the following are examples:

Removal of:	Special applications:
weld and nodular scale	gauging
slag	hydrostatic testing
rust and corrosion	swabbing (disinfectants, etc.)
lime scale	
ochre	
cement powder	
wax waste liquid	
sedimentation	
contamination	

Typical locations in which these commonly occur are:

Pipelines:	Process pipework:
crude oil	refined hydrocarbons
natural gas	petrochemicals
refined hydrocarbons	general chemical
chemical	cooling water
water	food and brewing
effluent	
dewatering	
slurry	

High-pressure water jetting

With high-pressure water-jetting techniques, deposits and blockages in pipes are broken up by the action of fine, high-velocity jets of water directed on to them from suitably shaped nozzles, and the resulting debris is washed out of the pipe by the flow of spent water.

Pressures of up to 1400 bar (20,000 lbf/in²), or in some cases higher, and a wide range of flow rates are used, depending on the nature of the material to be removed. Water is delivered to the nozzle through lances for cleaning small-bore pipes and tubes, and through flexible, self-propelling hoses for cleaning larger pipes.

With lances, straight pipes of 16 to 150 mm (⁵/₈ to 6 in) bore can be cleaned for a distance of up to 6 m (20 ft) from access points. Hoses are capable of cleaning pipes of 50 to 1020 mm (2 to 42 in) bore (up to 1.5 m (60 in) with attachments) for a distance of up to 120 m (400 ft) from access points, and they may also negotiate a number of bends.

Equipment

A conventional pump assembly consists of a diesel or electric power unit, high-pressure water pump, header tank, control valves and instrumentation, all mounted on either a skid, trailer or vehicle. Fully flame-proofed versions are available for use in hazardous environments.

The choice of pump is determined by the pressure and flow requirements of the pipe to be cleaned. Pumps are usually of the triple plunger type, enabling a range of pressures and flow regimes to be selected by changing plungers. The alternative diaphragm type of pump, often used in low-power applications, may also be used over an extended range of pressures, although this can only be achieved efficiently by a change of pump heads.

The power requirements of the pump are determined primarily by the pressure and flow rate. A useful rule of thumb is that, within the limits of flow and pressures for a given pump, the product of available flow rate and pressure are approximately constant. The majority of high-pressure water-jetting pumps are rated at between 25 and 200 hp, going up to 500 hp. A typical specification for a 100 hp pump, for example, would be to deliver 45 l/min at 690 bar (10 gal/min at 10,000 lbf/in²).

Hoses are selected to withstand the maximum operating pressure with an acceptable margin of safety, and to transmit the required flow rate with the minimum of pressure drop along lengths of up to 120 m (400 ft).

A primary consideration in the selection of lances for use in small pipes is the bore of the pipe, as sufficient clearance must be left around the outside of the lance without restricting the flow through the bore. Maximum lance length is 6 m (20 ft).

A range of nozzles is available to suit the application with a choice of the number and orientation of jets, and materials of construction.

High-pressure water is extremely dangerous. All pressure devices must be able to withstand the operating conditions, and must be properly maintained. Similarly, all operators must be correctly equipped with protective clothing.

Optimum cleaning methods

Careful selection of pressures, flow rates and nozzle configurations is an essential step if pipes are to be cleaned efficiently, and this depends upon possessing adequate information on the following factors:

- internal pipe diameter
- distance from access points
- number and location of bends and branches
- location of valves and other obstructions
- physical properties of material to be removed
- distribution and thickness of material to be removed
- health hazards

Table 7.

Pressure range	Example
Up to 275 bar (4000 lbf/in ²)	Grease, paraffin wax, crude residues algae, pulp, asbestos, PVA, food residues, loose masonry, clay, mud, silt.
275–550 bar (4000–8000 lbf/in ²)	Boiler scale, carbon, potash, asphalt, cement, plaster, mastic, PVC, unbonded paint, rust, oils.
550–1380 bar (8000–20,000 lbf/in ²)	Polymer, bonded paint, resins, plastics, synthetic rubber, coke, concrete, silicates, mill scale.

The physical properties of the material to be removed form the principal factor in determining the pressure required. Table illustrates cutting abilities of various pressures, although there can be considerable variations in practice.

Increasing flow rate has the general effect of increasing the rate at which material is cut. When large quantities of contamination have to be removed from a pipe, and particularly when it is completely blocked, high flow rates are essential. This is also true in the case of large-diameter pipes when sufficient cleansing flow must be maintained in the bottom of the pipe to keep debris in suspension.

The distribution of the material within the pipe influences the configuration of nozzle jets which are required, forward-facing nozzles being used to clear blockages, and rear-facing nozzles for flushing out loose material.

Applications

Water jetting is particularly suitable for removing large amounts of unwanted materials from pipes, including the cleaning of long lengths of totally blocked pipes. It offers advantages in plant downtime, cost and, very often, effectiveness of cleaning.

Pipe Cutting and Bending

Large-diameter iron and steel pipes are normally cut by mechanical saws or oxygen (flame) cutters in the manufacturer's shop. Special cutting machines are used for accurate production of profiles for pipe branchwork, saddles and square cuts.

Pipe-cutting machines and tools for general or on-site use range in size from powered tools capable of handling the largest sizes of pipes down to hand-operated cutters. The actual cutters may be circular saws (or cutting discs), reciprocating saws, wheels or knife edges. Hacksaws may also be used for hand cutting smaller pipes. A power-driven abrasive disc is one of the most widely used methods for cutting ductile-iron pipe of all sizes.

Circular-saw cutters are usually robust bench- or stand-mounted machines, but differ appreciably in detail design. Some employ milling cutter wheels, others conventional circular-saw blades. The majority encircle the pipe to be cut in a self-centring vice, but some are designed to travel around the pipe as they make the cut.

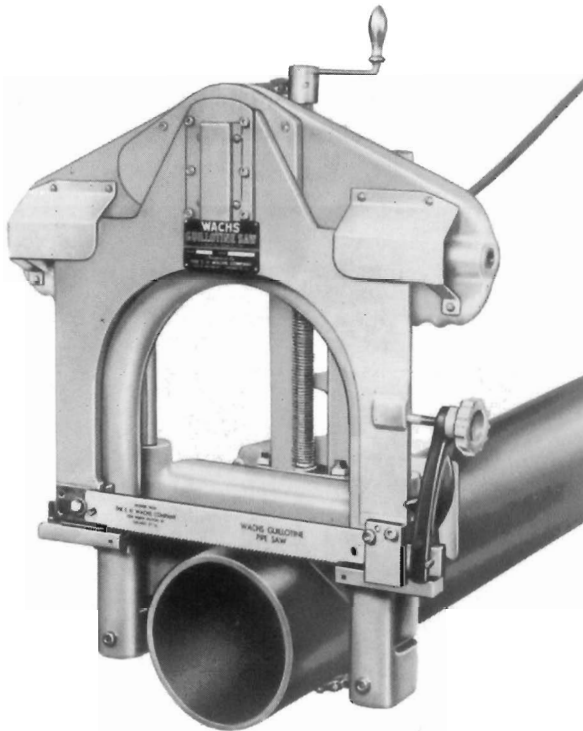


Power-driven abrasive disc cutter used for cutting ductile-iron pipe.

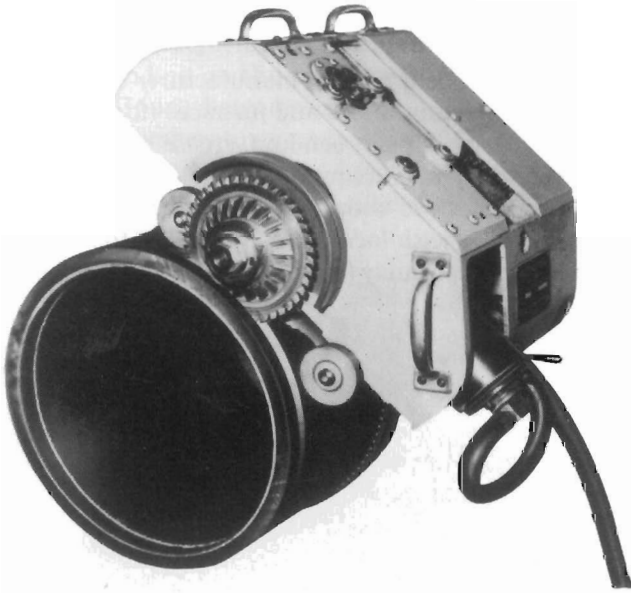
Reciprocating saw cutters range from simple machines with minimal guidance to guillotine saws where the tool is clamped to the pipe and the horizontal saw blade is fully guided to ensure a square cut.

Wheel or 'knife-edge' cutters are designed to encircle the pipe and hold it concentrically as the cutter is rotated around the pipe. On larger, power-operated machines of this type, the actual cutters are symmetrically positioned on a ring which is rotated by power. Each cutter is held in a tool box which automatically increases the depth of cut with rotation of the carrier ring and is profiled to remove equal amounts from the cut simultaneously.

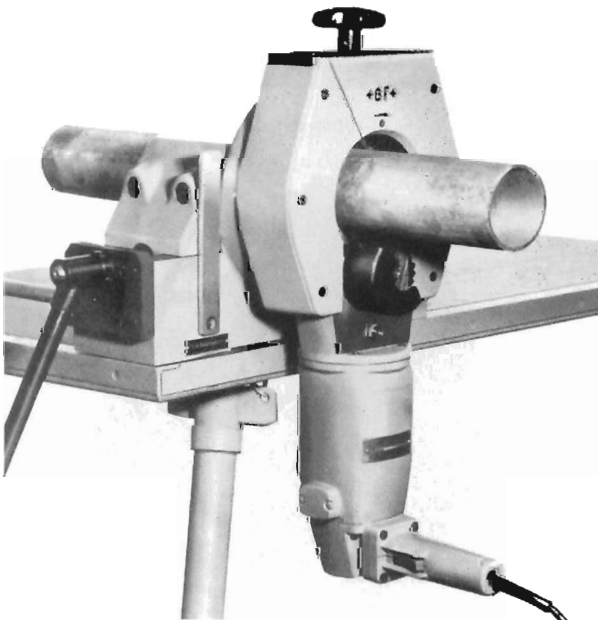
Manually-operated cutters normally employ cutting wheels rather than knife-edges, with a similar working principle (i.e. the tool is rotated around the pipe to produce the cut). The number of wheels employed may range from one to four. Cutter advance is by rotation of the handle (i.e. is not automatic). The other major difference is that the cutter does not have to be rotated continuously but can be 'rocked' backwards and forwards to make the cut if it incorporates three or four cutting wheels. This is a distinct advantage for close field work as only a relatively small clearance is then needed around the pipe being cut.



Guillotine pipe saw.



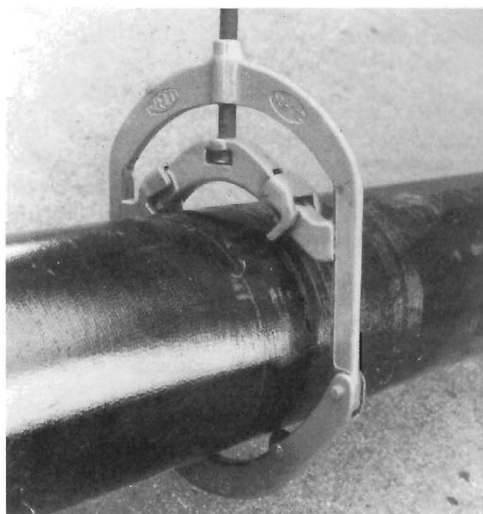
Pipe-cutting and machining tool.



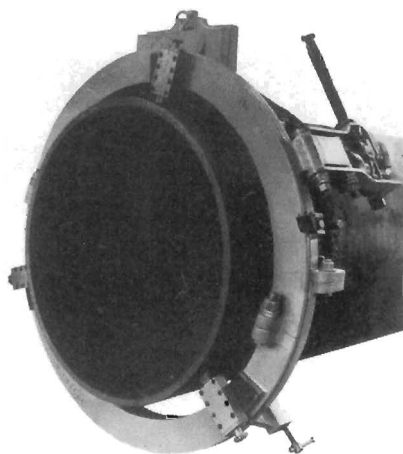
Electric pipe-cutting machine.

Pipe bends

Large-diameter pipes are bent in the manufacturer's pipe-bending shops using various types of bending tables and furnaces (for hot bending). Three main types of bends are used: plain bending, crease bending and corrugated bending. The former two may require the pipes to be filled with sand. Corrugated bends are produced with the pipe first corrugated in the straight on a corrugating machine with local heating and are then bent empty with each corrugation heated separately (e.g. by portable furnaces).

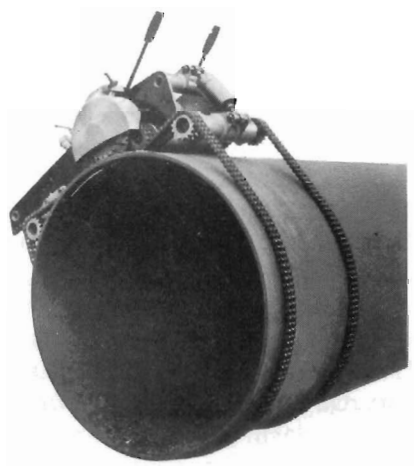


Semi-rotary wheel cutter.

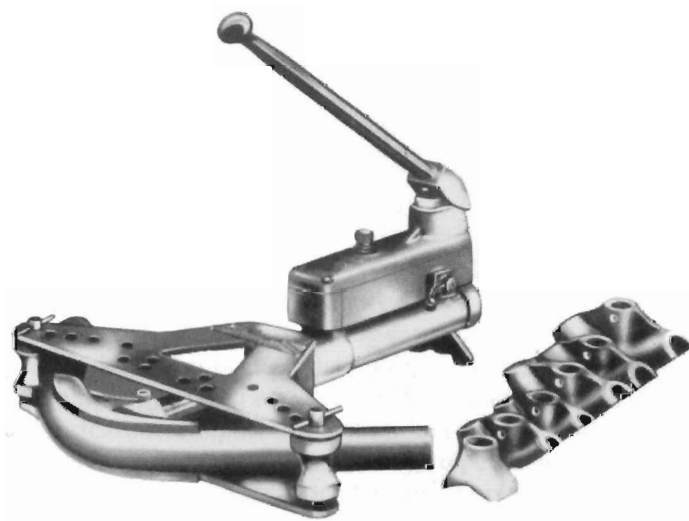


Rotary pipe cutter.

Plain pipes from 12 to 300 mm (2 to 12 in) may be bent cold on hydraulically-operated cold-bending machines, with bends up to 180° possible on radii depending on the pipe diameter and wall thickness. Plain pipes up to 900 mm (36 in) diameter or more can be bent hot, depending on the furnace capacity. Temperatures up to 1100°C (2000°F) may be used and heating time depends on the pipe size, type of bend and pipe material. For bending, the heated pipe is pegged to the bending table at one end and the other end pulled by a winch.



Orbital pipe cutter.



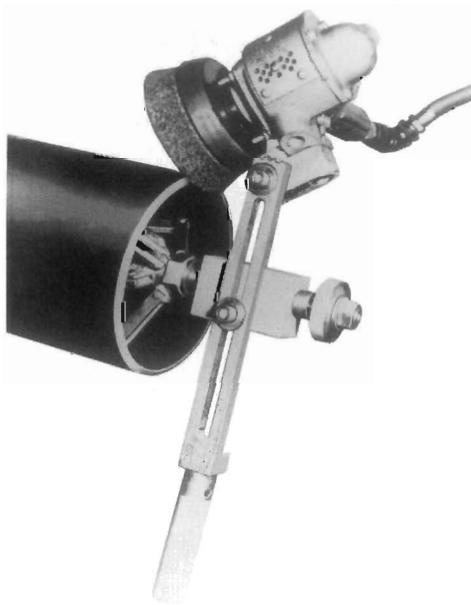
Hydraulically-operated pipe-bending machine.

The bending operator, who works to a curved template conforming accurately to the shape of the bend taken directly from a floor drawing, controls the rate of bending and the contour by using a coolant on the exterior of the pipe to control the heat spread. Water is normally used for low carbon steels, but is not recommended for alloy steels. Alternatively, stop pegs are used for control instead of a coolant; these are removable pegs inserted into holes in the bending table as required during the pulling operation.

After bending and allowing to cool, the pipe is emptied of sand, dressed and examined, and then checked on a surface table against a full-scale drawing. In all instances of hot bending, a great deal depends upon the skill of the pipe bender; movement may take place during cooling which must be predicted and allowed for, and the whole procedure of hot bending a large pipe calls for many years of accumulated experience for which there is no substitute. With thinner pipes, subsequent dressing with a flatter may be necessary to take care of slight rippling while the possible ovality of a pipe caused in bending is always under careful control. Sometimes, a resetting may be necessary to correct an error, for which gas-fired portable furnaces using premixed gas and air supplies are used.

Tube bending

Pipe bending by hand is only practicable in the smaller sizes of tubes, and then is not always satisfactory. On larger sizes, or with tubing with thin walls, it is



Bevel-grinding machine for pipes.

difficult to prevent local collapse of the inner wall unless a pipe bender or filler is used. Provided the tube material is reasonably ductile, all such bends are made cold.

The general rule for a minimum radius of bend is that this should not be smaller than three times the o.d. of the tube. A more generous figure is to be preferred, particularly in the case of the smaller sizes which are usually hand bent. Bent radii larger than the minimum values should always be used as far as possible because these produce less frictional loss and are less liable to result in deformation of the pipe section through wrinkling or stretching or introducing ovality. The latter is a common fault, even with pipe benders, unless extreme care is taken, and can materially reduce the working strength of the tube at the bend.

The four basic methods of machine bending are:

- (i) Press bending—particularly adapted to the bending of heavy gauge wall tubing up to six times the tube o.d. radii with included bend angles of 120° maximum. It can also be used with wing dies to achieve a minimum bend radius of three times the tube o.d.
- (ii) Roll bending—also suitable for heavy-gauge wall tubing and capable of achieving a satisfactory bend radius down to six times the tube o.d. The bend angle is unlimited because by using three power-driven rolls the tube can be fed continuously through the machine to produce complete coils. Four-roll bending machines are capable of producing true arcs right to the extreme end of the tube.
- (iii) Stationary die—a simple and popular method of bending smaller diameter sizes, usually up to about 16 mm ($5/8$ in) o.d. The tube is simply wrapped or 'whipped' around a grooved bending die. Both circular and non-circular bends can be produced, also bends in two planes (using special dies). This method is, therefore, extremely versatile.
- (iv) Revolving die—in this case the die is rotated while the bending shoe remains stationary. The particular advantage of this method is that the tubing can be entirely confined internally and externally at the point of bend, thus minimising the risk of distortion. The revolving die machine is particularly suited to handling thin-walled tubing.

Proprietary pipe benders are usually based on one or other of these methods. All aim at producing the bends down to a specified minimum radius with minimum distortion of the tube material. Many, it will be appreciated, involve a 'wiping' or rubbing action over the outer surface of the tube and in such cases lubrication is important. Light mineral oil is a satisfactory lubricant for bending steel tubes. Some designs of pipe benders are designed to compensate for 'thinning' effects on the outside wall of the formed bend so that the distribution of material over the cross-section remains unaltered. Others may produce appreciable thinning.

Instead of pipe benders, filler may be used to support the inside of pipes and tubes for manual manipulation, or even be used with pipe benders to prevent distortion. The best type of filler for the purpose is a low melting point alloy which can be removed by gently heating after the bend is completed. The use of low melting point metallic fillers is not, however, generally recommended for bending hydraulic tubes as it is difficult to remove completely all traces of the metal. The only effective way of ensuring complete removal is usually to blow through the pipes with steam. Sand is not generally used as a filler as it is difficult to ensure its complete removal after forming, unless elaborate pressure-cleaning methods are used.

The quality of the bend produced, whether manipulated by hand or machine, is very much dependent on the operator. Jerky or irregular actions may produce kinks or wrinkles. Wrinkling may also occur on the inside of the bend due to the compression of the material in this region, unless the machine compensates for this by applying tension to the inner radius.

Thickness of bends

The minimum thickness (tb) of a straight pipe from which a pipe bend to a radius in accordance with Table 1 is to be made shall be determined from

Table 1. Minimum bending radii for pipes of thickness determined by BS formulae

o.d. mm	Radii measured to centre line of pipe	
	tb = 1.125 tf all thicknesses mm	tb = 1.1 tf tb = 35 mm or above mm
26.9	65	
33.7	75	
42.4	100	
48.3	115	
60.3	150	
76.1	190	
88.9	230	
101.6	265	
114.3	305	
139.7	380	
168.3	460	
193.7	630	
219.1	710	
244.5	810	1140
273.0	1020	1270
323.9	1220	1520
355.6	1500	1780
406.4	1730	2030
457.0	2030	2080

equation (i) or equation (ii), except where it can be demonstrated that the use of a thickness less than t_b would not reduce the thickness below t_f at any point after bending.

For pipes 219.1 mm o.d. and below, and for pipes above 219.1 mm o.d. bent to the radii specified in the table, column 2:

$$(i) \quad t_b = 1.125 t_f$$

For pipes above 219.1 mm o.d. where t_f is 32 mm or more, bent to the radii specified in the table, column 3:

$$(ii) \quad t_b = 1.1 t_f$$

The value of t_b is the minimum thickness and provision shall be made for minus tolerances. Manufacturing considerations may make it necessary for pipes thicker than this minimum to be used.

Radii of bends

Pipes complying with the requirements of BS 1387 and BS 3601 shall not be bent to radii less than those given in Table 1. Other pipes of a thickness determined by BS formulae shall not be bent to radii less than those given in Table 1 unless:

- (a) It can be demonstrated that the use of this thickness will not reduce the thickness at any point after bending to below t_f , and
- (b) where the design temperature of the piping is higher than 430°C in the case of alloy steels and the radius is less than three times the i.d. it can be additionally demonstrated that the thickness at the internal radius of the bend is not less than resulting from the following:

$$t_i \geq t_f \frac{2R - r}{2R - 2r}$$

where

t_i is the thickness at internal radius (mm);

R is the radius of the bend (mm);

r is the mean radius of the pipe (mm).

In general it will be necessary to increase the thickness above that determined by BS formulae in order to meet the aforementioned requirements.

There is a minimum thickness for each size of pipe, depending on bending procedure, below which the allowance for thinning will be exceeded and, in such cases, the radius given in Table 1 should be increased where necessary to ensure that the thickness is not below t_f at any point after bending.

Rigid thermoplastic pipe

Rigid PVC and CPVC plastic pipe can be readily cut with an ordinary hacksaw or power saw. A cutting speed of 6000 r/min using ordinary hand pressure is recommended. With band saws, a cutting speed of 3600 ft/min using hand pressure is recommended. Under some circumstances a lathe can be used. Best results are obtained with fine-toothed saw blades (16–18 teeth per inch) and little or no set (max 0.025 in).

Cuts should be square and smooth, particularly if the pipe is to be threaded. A mitre box or similar guide should be used when cutting by hand. The cut ends can be bevelled with a hand file and the interior deburred with a regular tool or knife. Dust and chips should be removed to prevent fluid-stream contamination. The pipe should be well supported during cutting and protected from nicks and scratches by wrapping in canvas or similar material.

Use of wheel-type pipe cutters is not generally recommended since they tend to generate heat and can produce a raised bead or ridge which increases the bevelling effort required.

Bending may sometimes be advantageous in fabricating PVC and CPVC pipelines. However, bending should be limited to non-critical applications at room temperature or lower where maximum operating pressures are not utilised. With the procedure normally used in bending, some stresses from bending are retained in the material in addition to those caused by the pressure of the medium.

If bending has to be done, the pipe should be heated from 120 to 135°C (250 to 275°F) by use of a flameless hot-gas torch, hot-air oven, or by immersion in hot oil. Uniform heat distribution is required and localised overheating must be avoided. Care should be taken to avoid holding the pipe at bending temperature for too long as the pipe may lose its form. The pipe should be bent around a regular pipe bending form of the required radius, grooved to the proper diameter and having a radius at bend not less than five times the pipe outside diameter (to prevent flattening). Other proven methods include filling the pipe with sand or the insertion of a coiled pipe spring before bending.

Because of the recovery characteristics of the pipe, it should be bent slightly beyond the desired radius and allowed to spring back, then quickly cooled in water or with air. It is recommended that the pipe manufacturer or supplier be consulted regarding the bending suitability of plastic pipe. Thermoplastic piping is a general term applied to a variety of different plastics.

Hot tapping and plugging

Hot tapping is the procedure for cutting an opening into cast-iron, ductile-iron and steel pipe which is carrying a product under pressure. A fitting is welded or mechanically attached to the line and a valve is attached to the fitting. A tapping machine is installed and the tap made through the valve.

After the tapping is made, the cutter is withdrawn and the valve closed. If a completion plug is to be installed in the fitting, the valve can then be removed.

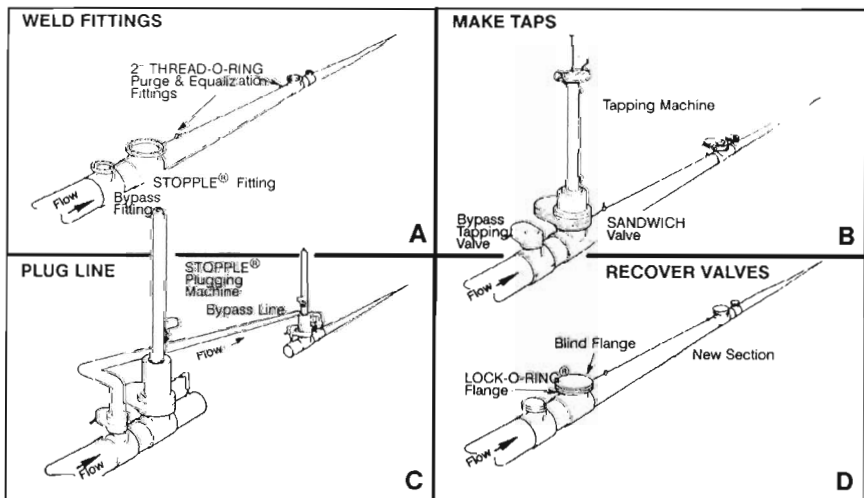
In the petrochemical industry, for example, it is often necessary to isolate a section of piping without interrupting the service the line provides. Lines must be kept in service to avoid the shutdown of an entire unit. Hot tapping and plugging equipment is designed to meet these requirements.

Plugging occurs after the tap is made. Typically a STOPPLE™ plugging machine is installed on the valve and a plugging head inserted into the line. The plugging head serves as a block valve and seals the line retaining the pipeline pressure.

If two plugging machines are used, or one plugging machine and an existing in-line valve, a section of pipe can be isolated and drained, making possible necessary repairs or modifications in the isolated section. A bypass can be installed around the isolated section, keeping the line in service. If a new section of line is being installed, it is possible to use the new section for a bypass while the old section is being removed. When all repairs have been made, the job is completed by installing a completion plug in the fitting.

The plugging head is removed, restoring service through the line. The tapping machine is then fitted with the completion plug, inserted and locked into the fitting to provide the seal. The tapping machine, bypass and tapping valve are removed and a blind flange installed on the fitting. In most cases the blind flange can be removed and the line re-entered at a future date after the completion plug has been removed.

One of the most common hot tapping and plugging applications is used when a valve in a piping system no longer works properly or is damaged.



Hot tapping and plugging sequence.

If it is not necessary to keep the line in service, the line can be tapped and plugged upstream of the valve using only one plugging machine.

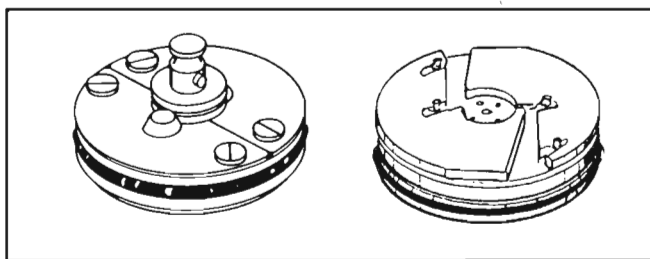
Where it is necessary to keep the line in service while repairs are made, two plugging machines are set, isolating the faulty valve. The use of a three-way T or an adapter with a side outlet eliminates the need for additional tops.

In plugging many types of product lines, slight seepage across the plugging head can sometimes occur. In the case of hazardous products, a better seal must be obtained with the plugging head before the work is commenced.

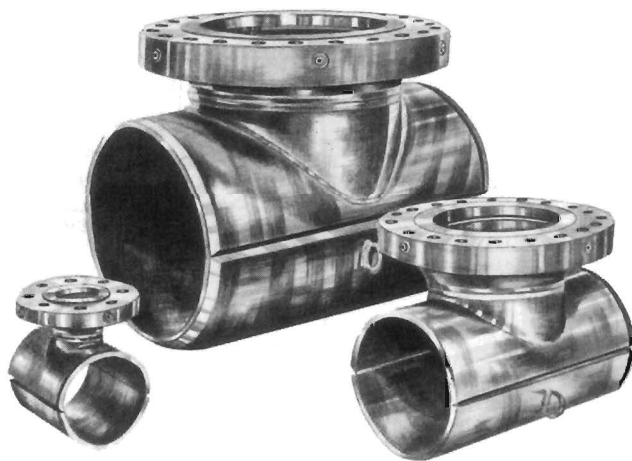
Folding plugging head

This concept is used on low-pressure lines (maximum 150 lb/in^2 depending on the size of the line) to plug a line through a reduced branch fitting.

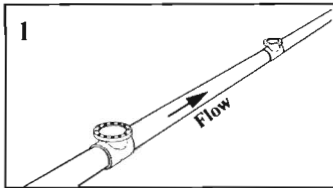
The folding plugging head involves a flexible sealing element which is attached to a plugging head with hinged leaves. Depending on the size, the leaves are opened in the line either hydraulically or mechanically.



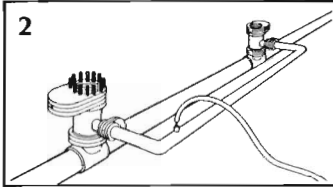
Typical completion plugs.



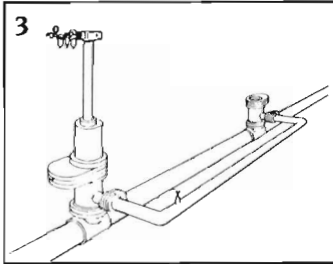
High-pressure STOPPLE™ and tapping fittings.



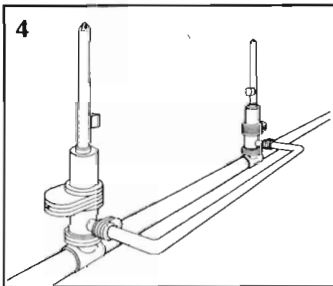
First, two STOPPLE® Fittings are installed on the pipe.



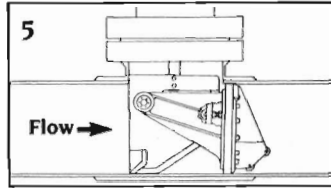
The temporary bypass is installed and pressure tested.



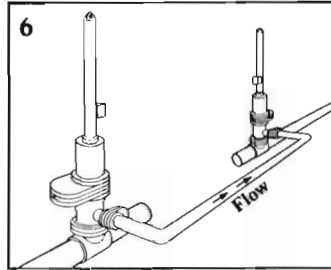
The tapping machine makes taps through both STOPPLE Fittings.



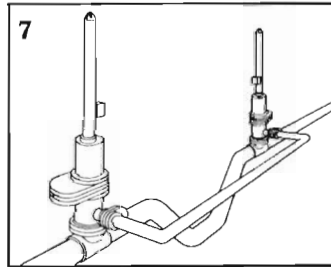
The STOPPLE® Plugging Machines are installed.



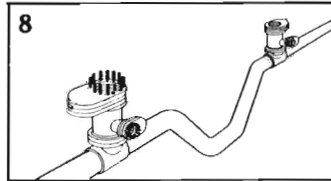
The plugging heads are lowered into sealing position, diverting flow through the temporary bypass.



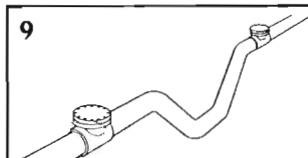
With all flow diverted through the bypass, workers cut and remove the isolated section.



Once the new section is tied in, the plugging heads are retracted, returning full flow to the main.



The plugging machines and bypass lines are removed.



Completion plugs are set in the STOPPLE® Fittings and the SANDWICH® Valves and spools are recovered, completing the job.

Typical example of a tapping and plugging application using a temporary bypass to maintain flow.



The 96-in plugging head, in its folded position, is retracted into the 60-in housing. It is lowered through the tapped hole into the pipe. When the guide wheel, shown at the top, touches the bottom of the pipe, the plugging head opens and the elastomer sealing element seals against the wall of the pipe, assisted by differential pressure.

Welded fittings

Extreme care must be taken when welding onto an in-service pipeline.

Two major concerns are:

1. Burn-through—where the pipe wall is penetrated allowing the contents to escape.
2. Hydrogen cracking—as a result of high hardness levels from the accelerated cooling rate associated with the ability of the flowing pipeline contents to remove heat from the pipe wall.

A thorough understanding of factors related to welding and, in particular, to welding on in-service pipelines is required to ensure safe operating procedures and sound welded joints.

For some applications, the heat input required to avoid cracking may be greater than the heat input allowed to avoid burn-through, making welding prohibitive.

In addition to hot tapping and plugging cast-iron, ductile-iron and internally-coated steel pipe, it is also possible for tapping and plugging to be accomplished on most reinforced-concrete pipe, provided it is done within the pressure limits required when the steel shell is exposed to mount the fitting.

Pipeline Inspection and Evaluation

Pipelines are normally surveyed for one of the following reasons:

- (i) To investigate repeated or isolated blockages.
- (ii) To check on structural integrity, for connections from new sites, or expected working life for maintenance expenditure budgets.
- (iii) To determine exact location of pipelines, branches and connections to update drawings or carry out maintenance work.
- (iv) To inspect installations prior to hand-over.
- (v) To check on scale or corrosion build-up on industrial heating or process pipelines.
- (vi) To detect and locate leaks in buried pipelines.
- (vii) To carry out interior inspection of welded joints on long rungs of metal pipe (e.g. gas or oil lines).
- (viii) To test individual components, e.g. safety and relief valves on line.

Techniques used obviously vary with individual industries and installations, size of pipes, etc.

Defects in pipelines cannot be effectively evaluated thoroughly without some form of inspection, both internal and external, being undertaken. Many systems have been engineered and developed for this purpose.

Public authorities and municipalities have been recording video tapes of the activity in their pipelines for over 30 years, together with corresponding inspection logs which have caused massive retrieval problems. Hardware and software is now available to catalogue and reduce mountains of material into a highly effective data collection and information retrieval system.

Inspection data normally include four basic materials: video tape, inspection log, defect classification/cataloguing and picture capture.

Video tapes provide a good visual picture of a pipeline's overall condition. Defects, though, will only generally represent a few seconds in a tape lasting from one to one and a half hours. If an inspection log is available, the user can fast-forward the video tape to the exact point of the tape. If no inspection log is available, then the tape has to be scanned continuously to locate the defect.

Inspection logs can vary considerably in their content. A good log should be identified to a corresponding video tape, with both tapes and logs indexed and referenced as to exact location.

Some logs contain much more information than just defect data and location. More sophisticated logs incorporate contract information, line location, pipe size, pipe type, line gradient (or slope) and other conditions.

Logs are typically stored on a reproducible medium such as a floppy disc. Software is available to classify and catalogue defects by type and severity of defect. These programs allow the automatic sorting of defects in order of severity and rehabilitation requirements. This information then serves for scheduling repairs and budgeting projected costs.

For a classification program to be effective, continuity in defect classification is mandatory. When a single operator performs defect classification, there is less of a problem than when different crews, operators or contractors perform inspections and defect classification. Photographs of each defect or problem should be taken and, if possible, this should be carried out using digital equipment to avoid the possibility of picture degradation due to ageing. A digital picture will not fade or degrade with age, humidity or temperature and can undergo enhancement and reproduction without much limitation, using a standard computer and ink-jet or similar colour printer.

Stand-alone inspection vehicles are particularly useful with municipal waste-water collection systems. They typically have a data system that will measure the pipeline gradient (slope), label a defect, insert its distance location (distance from entry) and label an operator-selected defect classification. This alphanumeric data will display simultaneously on the video monitor and be recorded on the video tape and stored on a magnetic disk.

The addition of a graphics computer and picture-capture software makes it possible to store a still-frame digital picture of a defect or other significant item simultaneously on a magnetic disk. It is possible to copy all the data including the captured photograph to a removable disk for integration into a master database or geographical information system (GIS). Inspection reports and hard-copy photographs of captured defects should be placed in a master defect log book.

Information and data from each collection vehicle or base should then be ultimately stored in a central library to provide easy access to all available historical data.

Geographical information systems

Geographical information systems (GIS) are used to collect and store information on pipelines and tunnels over large area networks. GIS databases now contain information on pipeline inspections and surveys using CCTV equipment. When a survey is completed for a particular pipeline or section of pipeline, the information is stored in the database where it can be accessed quickly and assessed via on-line/off-line multi-media tools.

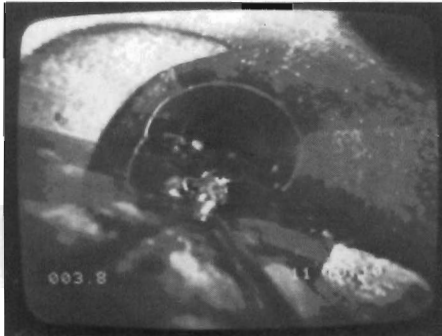
TV surveying

Control is the key to inspection systems and TV surveying is a particularly versatile method of pipeline inspection, generally applicable to pipes ranging in size from 50 mm to over 1 m (2 in to over 4 ft) in diameter. Such systems can also be designed to function under widely varying conditions. The following will be required: video camera, umbilical cable, a control unit, lighting leads, a video signal recorder (e.g. videotape), power supply and ancillary equipment.

Video camera units may range from as small as 25 mm (1 in) diameter upwards. (Small units are obviously necessary to survey small-diameter pipelines and negotiate bends.) A normal 25 mm (1 in) diameter camera, for example, can be expected to negotiate 90° bends in 100 mm (4 in) diameter pipelines.

Solid-state mini-cameras can now inspect pipe as small as 25 mm with 90° bends.

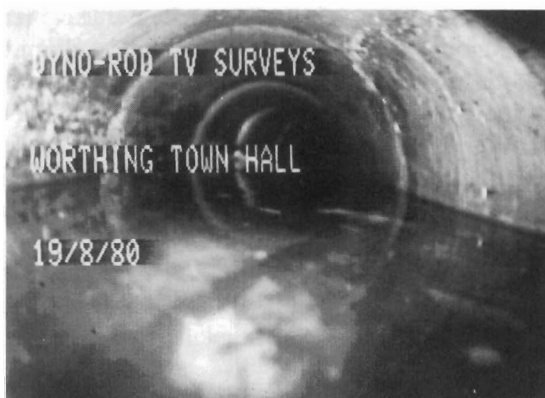
Sizes vary but, typically, camera heads the size of a golf ball have high intensity LED (light-emitting diode) bulbs and a lens with a 270° rotational viewing angle.



Built-in sondes transmit radio signals from within even cast-iron pipes to provide accurate location of the camera in the pipeline.

Sewer camera systems generally comprise a choice of camera size, 60, 90 or 120 m of flexible rod wound onto a rotating frame, a high-resolution VCR monitor and a locator. Power for the system is usually mains supply or 12 v d.c. for remote operation.

Various lighting attachments may be used for illumination of pipes of different diameters, including the use of a rotating mirror in front of the lens at



Typical pipeline inspection system.

90° to the line of the pipe so that ports and branch lines can be screened circumferentially.

An umbilical cable of suitable length connects the camera to the control console. This carries lighting power, rotating-mirror power and vision signal; it is stored on a 12-way slipping drum or similar device that contains a trip-measuring device plus an electronic pulsing unit which relays back to the monitor the distance the camera has travelled down any one line. This distance can be zeroed or preset to any measurement at any one time.

The control unit, which is basically the power unit for the camera, controls lighting intensity, remote focusing capability, and rotating-mirror positioning.

Lighting heads are normally low-voltage lamps of different sizes and wattages housed in intrinsically-safe, glass-fronted housings for attachment to the front of the camera.

Various items and relevant information need to be shown on the monitor screen during a survey, such as: site address, date, pipe diameter, run number, location and description of faults, etc.

The equipment used to produce the characters shown on the screen can come in various shapes and sizes with differing capabilities, but it is normally known as a word processor or screen writer.

Ancillary equipment carried by the unit may include: generator, winches, steel rods, drain stoppers, lifeline, harness, hydrant stand pipe, hoses and key, extension leads, gas detector, road cones, cable rollers, field telephone set with up to 365 m (400 yd) of cable float lines, etc.

Survey methods

A tried and tested method used to carry out a TV survey is that shown in Figure 1, whereby a line is passed through the run and a camera is towed by a winch from chamber 'A' to chamber 'B'. Winch 'A' is purely a safeguard, so that if the run being surveyed is damaged to such a degree that camera progress is impeded, the camera can be retrieved by winching back on a steel

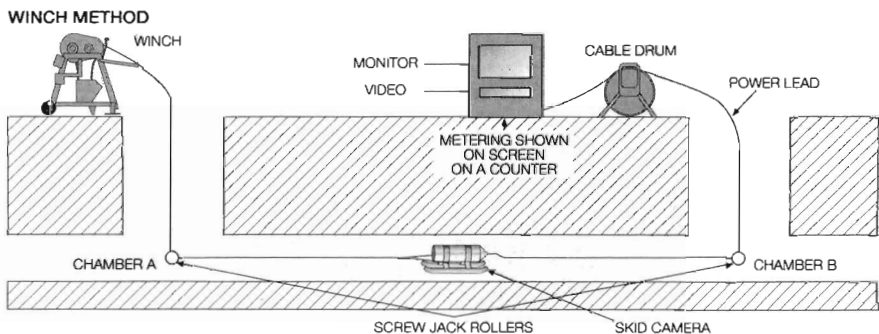


Figure 1.

band instead of pulling on the umbilical cable. As previously stated, as the camera and umbilical cable are pulled down the run, the measuring wheel on the drum transmits back to the monitor the distance the camera has travelled from the beginning of the run for location purposes.

Runs are normally surveyed in the direction of the fall of the drain. The reason for this is that if the camera is pulled against the flow it can create a bow wave, causing droplets of water to splash on to the camera lens, so distorting vision. Waste matter can also be deposited on the lens which could obliterate vision completely. Surveying against the fall is only possible if the run is 'dry' during the survey.

A landline telephone system may be used for communications between vision controller and winchman, so that the camera can be stopped anywhere along the run to focus on any problem area.

When the survey is completed the camera and skid are detached from the umbilical cord and safety winch cable. The two cables are then wound back to chamber 'A'. Other methods of transporting the TV camera are by rodding (shown in Figure 2) and self-propelled traction units.

Duct tractors are capable of providing a vehicle for tasks as varied as simple draw-line installation, the positioning of remote inspection devices such as sensors, as well as closed-circuit TV surveillance applications. One system uses a self-propelled small-bore pipe crawler with traction being created by the deformation of counteracting diaphragms on the internal walls of the pipe.

This method of traction allows the tool to operate along pipes running at various inclines, including vertical limbs of up to 100 m (328 ft). The crawler can also traverse holes in the pipe wall of up to 75% of the pipe diameter and, in the straight pipe condition, over holes of up to 100% of the pipe diameter, i.e. inverted equal-T junctions.

The energy source is compressed air or an inert gas, depending on environmental conditions, delivered at the appropriate pressure along an

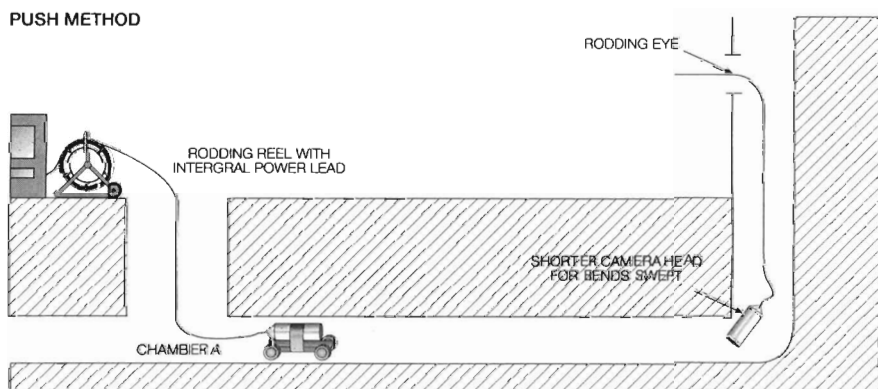


Figure 2.

umbilical supply line. The crawler is designed to operate in most dry, commercially available ductways with the exception of PTFE-lined pipes. In general, the rougher the traction surface the more satisfactory the operation. The vehicle will also travel at reduced speed through a pipe which at any point contains up to 20% non-viscous liquid, e.g. water.

Figure 3 shows a duct tractor unit complete with trailed pneumatic logic carriage, umbilical interface carriage and 50 m (164 ft) reel of drive supply line on drum.

Long pipelines and tunnels

Several techniques are used for inspection of long pipelines and tunnels. Varieties of autonomous underwater vehicles (AUV), ranging from the dumb pig swept through by water flow to smarter systems capable of recording their progress and the state of the pipeline or tunnel as they pass, operate with variable success rates. The problem has always been whether to have free-swimming vehicles in the pipeline or tunnel. Remotely operated vehicles (ROV) are a viable proposition if cable drag can be kept within acceptable limits. Good cable needs to be neutrally buoyant and carry sufficient copper and fibre-optic links. It also should be under 15 mm in diameter with a polyethylene outer jacket to provide a low coefficient of friction. ROVs operate on tethers of 10 km without problem. Remotely operated vehicles have

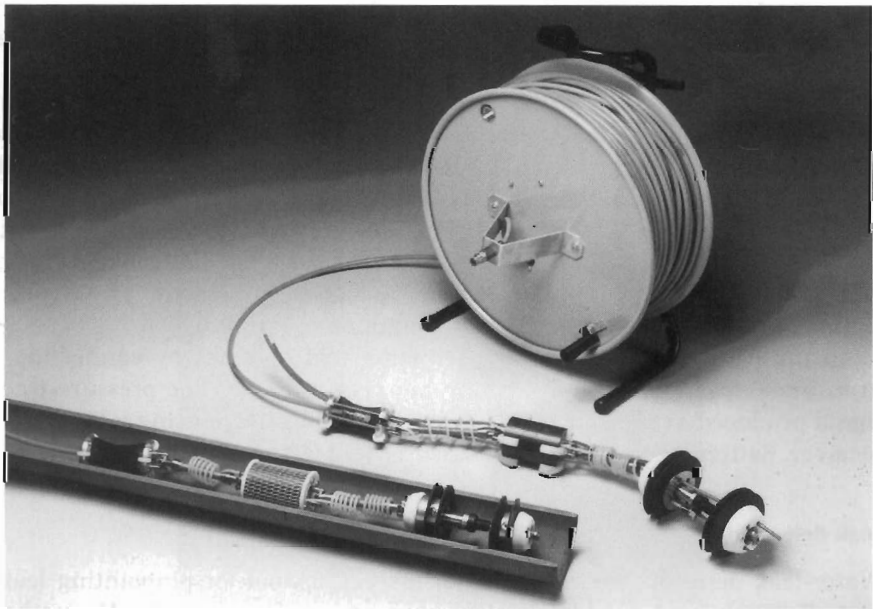


Figure 3. Remotely-controlled duct tractor capable of tackling severe bends in small-diameter pipes.

a number of thrusters, (usually 12, eight forward and two each for vertical and lateral movement).

Control software typically includes the standard diagnostic features and the unit is capable of handling up to four video cameras. In addition to those used for inspection work, one usually faces rearward to help vehicle retrieval.

Sonar fitted to the vehicle are used for collision avoidance and scanning (e.g. short-range profiling).

As the vehicle progresses down the pipeline or tunnel, it rotates and scans a helical path. The sonar screen displays the tunnel cross-section. Flattening at the bottom indicates sediment build-up and features elsewhere show either wall corrosion or a build-up of mineral deposits, usually around flanges. Two forward-looking cameras look at abnormalities thrown up by the sonar.

The biggest problem usually encountered by ROVs is the amount of debris in the water in spite of trash rakes and sieves. Plastic-material debris can snag the propellers in the thrusters and plastic bags or similar objects can become wrapped around the sonar head, for example. The chances of traversing a 10 km tunnel or pipeline without encountering any obstruction are probably very small.

Position surveys

Metal detection can be used to trace the path of buried metal pipelines, and also plastic pipelines where a metalised identification strip has been buried with the pipe. Simple hand-held units of this type are effective up to depths of 2 m (6 ft). More powerful units may be used for detection at greater depth, and also for surveying wrapped or coated pipes. In the latter case the instrument may also be capable of detecting flaws or breaks in the pipe wrap.

Such instruments may work in the inductive or conductive mode. In the inductive mode, a linear or circular aerial is used as a 'direction finder' to establish a small signal which identifies the exact location of the pipeline run relative to the aerial. With conductive mode operation, no aerial is used and direct contact is made with the pipe to be traced to feed the signal directly into it.

Cable-avoiding tools (CAT) and transmitters offer advantages in terms of accuracy and speed for pipe and cable location equipment. This type of equipment can be used in all weather conditions.

An innovative plastic water pipe tracing tool operates by connecting a 'transonde' transmitter to a running tap or stand pipe. The pressure-wave signal produced in the pipe can be located on the surface using a hand-held receiver. Battery-operated versions are used for tracing service pipes.

Leak detection

Water-leak detectors are an essential inspection tool for pinpointing leaks cost-effectively and quickly. Typical apparatus is based on the H_2 method where a low concentration of hydrogen is 'injected' into the leaking pipe and

the escaping gas is then located using a specially designed hydrogen-gas detector. The gas can permeate through most materials. It is a 5% mixture in nitrogen and therefore non-explosive in this diluted state.

Line protection

Line protection systems are pipeline monitoring and line-break detection devices controlled by a dedicated microcomputer. The system is normally located at a pipeline valve site to provide supervisory control of the valve and actuator.

Line protection systems continuously monitor the pipeline pressure at a point near the valve site. Once armed, and when abnormal pressure conditions are sensed, the protection system strokes the valve to a fail-safe position. The system operates in various modes including:

Data collection mode—pressure is sampled at 32 second intervals and 30 minutes of pressure history is stored in a temporary rolling memory. Memory capacities are usually quite large.

Valve control mode—pipeline pressure is sampled every 8 seconds. The pressure magnitude and rate of drop are continuously compared to the user-set values. When pre-set values are exceeded for the specified time duration, the control will cause the valve to stroke the fail-safe position. Internal relay contacts interfaced with telemetry or SCADA systems send a warning signal back to pipeline operations personnel.

Communications mode—used when the system is connected to a portable computer.



Water leak detector.

Location of sub-marine pipelines

Four different methods can be used to locate sub-marine pipelines: echo-sounding, magnetometers, sonar-scanning (and similar acoustic methods), and seismic profiling. Echo-sounding and acoustic methods can only be used to detect sub-marine pipes which are exposed above the seabed. Magnetometers and seismic systems can detect buried sub-marine pipelines.

The most common method of locating exposed sub-marine pipelines is by side-scan sonar, using a towed 'fish' with laterally-directed transducers generating acoustic pulses of frequencies of the order of 100 kHz. Various techniques are used depending on the depth of water involved, the length of pipeline, etc., and the feasibility of employing underwater acoustic transporters. In shallow waters, suitable results may be achieved more simply and economically using transit sonar, when the sonar transducer is mounted on the hull of the running vessel, or conventional high-resolution echo sounders.

The application of a magnetometer or seismic profiler for the detection of buried sub-marine pipelines requires the use of detectors towed behind the survey vessel which then performs a series of transverses to intersect the expected pipeline axis.

Magnetometers measure the strength of the earth's magnetic field, which is affected by the presence of a steel pipeline. Essentially, then, a magnetometer detects the presence of such a buried pipeline by its anomalous magnetic effect. This magnetic effect is usually proportional to the centre of the distance



Linebreak protection system.

between the magnetometer sensor and the object (pipeline). For this reason it is necessary to lower the magnetometer 'fish' as close as possible to the seabed.

With a seismic profiler, a piezoelectric transducer or 'pinger' emits pulses of acoustic energy with single frequencies in the range 1.25 to 1.4 kHz at a high pulse rate. The 'fish' is towed at a height of 5 to 10 m (15 to 30 ft) above the seabed, concentrating the downward-directed energy beam over a small area of seabed. A proportion of acoustic energy penetrating the seabed is reflected back by the lined pipeline, producing a characteristic deflection on a graphic recorder.

Jacketing and Dual Containment

Even the most engineered piping system is subject to temperature-control and containment problems. Process industries require higher and more precise temperature control for efficient operation.

A range of thermal-jacketed products are available for organic and inorganic chemical processors, pharmaceutical plants, polymer producers, petrochemical plants and food processors. These products cover a broad range of processing components including pipes, valves, strainers, fittings and pumps.

Generally, jacketing that has been specifically fabricated falls into one of three broad categories: standard, swaged and hybrid systems.

Standard jacketing

Typically, this system provides uniform application of heat by covering the pipe or valve (core) from the flange. The jacket is welded to the back of the flange so oversize valves must be used to accommodate bolts.

Figure 1 shows the standard jacketing system.

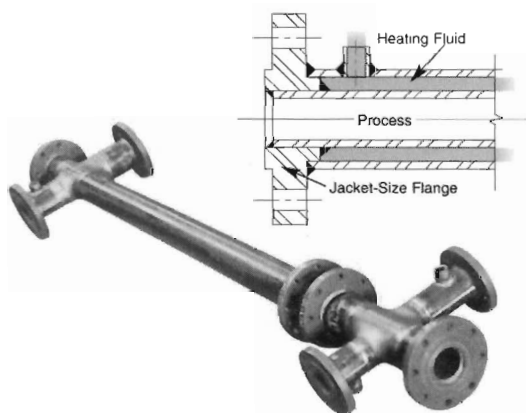


Figure 1. Standard jacketed pipe.

Swaged jacketing

Also referred to as capped or partial jacketing, this system is often used where protection against cross-contamination is required and where temperature discontinuities at flanges can be tolerated. Swaged jacketing can be less expensive than standard jacketing because small in-line flanges can be used (Figure 2).

Hybrid jacketing

This method utilises a combination of both swaged and standard jacketing systems as well as removal and special jacketing. Straight-line piping may use swaged jacketing while valves and fittings employ standard or removable jacketing to eliminate temperature discontinuities at critical flow areas (Figure 3).



Figure 2. Swaged jacketed pipe spool with stainless core and carbon jacket. The jacket has an integral stainless expansion joint to relieve cyclic heat stress.

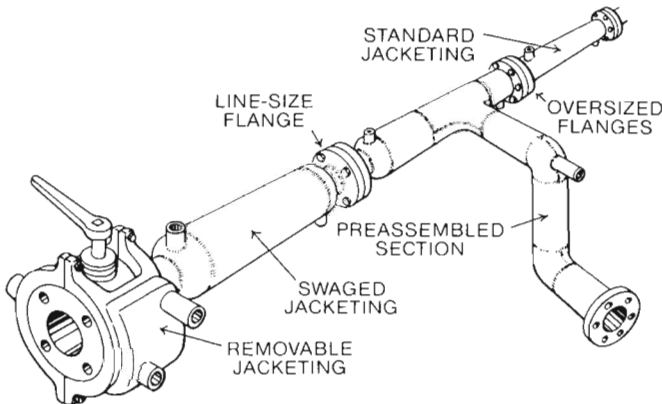


Figure 3. Typical hybrid jacketing.

Jacketed system design

Very little information is readily available regarding design considerations and recommendations for jacketed piping systems.

Fabricated jacket equipment manufacturers are the best placed to provide the information as a result of their manufacturing and fabricating experience and accepted practices within the processing industries. A typical manufacturer's recommendation to be considered in overall system design is given here:

1. Design for uniform temperature control throughout the system. Chill spots, even in moderate temperature systems (120–180°C; 250–350°F), are the problems most frequently encountered. Temperature discontinuities at flanged connections, valves or fittings may cause product build-up and solidification at critical points.
2. Design for uniform heat stress. This is particularly important where temperatures are over 120°C (250°F) and in batch-type service where the piping is subject to frequent heat cycles. Under these conditions the core and the jacket should be of the same material or have very similar coefficients of expansion.
3. Lengths of straight-jacketed pipe should be a maximum of 6 m (20 ft) for ease of installation.
4. Spacers between the core and the jacket on pipes should have a nominal clearance of 1.58 mm ($1/16$ in) to restrict core sag while allowing differential motion caused by heat stress.
5. The slope of installed jacket pipe should be gradual, about 3 mm per 300 mm ($1/8$ in per foot) to eliminate pockets and facilitate drainage of the heating fluid from the jacket.
6. Heated fluid should flow counter-current to the process to promote the most uniform application of heat.
7. The length of jacketed runs or the number of spools per single supply of heating fluid should be carefully analysed. Almost every single application will have widely varying parameters.
8. Jacket jumpovers, depending on the size of the jacketed pipe and the heating throughput required, should be 25, 19 or 12 mm (1, $3/4$ or $1/2$ in). Flexible metal-hose jumpovers can be used with both vapour and liquid heating media. System pressures determine whether the hose should be braided or unbraided. In certain high-temperature, high-pressure applications, pre-formed metal tubing may be preferred to metal hose.
9. Metal selection for jacketed piping usually has two major considerations: performance and cost or performance versus cost. The most frequently used jacketed-piping systems by the chemical process industry is standard weight carbon steel (SA53) for both the core and jacket. Food and pharmaceutical processors generally specify stainless on stainless construction for service above 120°C (250°F), and carbon on stainless construction for lower-temperature services.

Fabricated jacketed assemblies

Fabricated assemblies provide considerable savings on material costs and save on installation time. The assemblies are easier to insulate, with few jumpovers and fewer flanges.

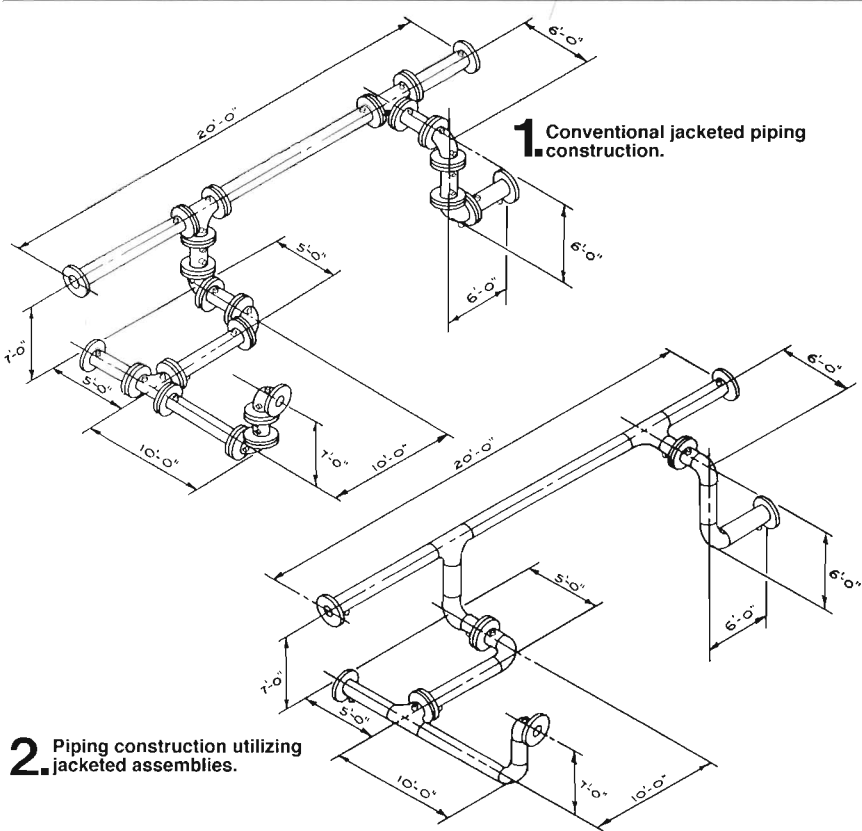
Table 1 compares a conventional jacketed-piping construction with a piping construction utilising jacketed assemblies, both using the same piping section.

Jacketed valves

Practically all types of valves can be fully jacketed by fabricating techniques, including many valves not available with integrally-cast jackets (Figure 4).

Standard fabrication includes modifying the valve to accept oversize flanges, extending the body as necessary to ANSI standard, then adding the

Table 1. Conventional jacketed-piping construction versus jacketed assemblies



full jacket ensuring that the interior tolerances remain the same as the original unjacketed valve. Typically, there are four basic types of jacketed valve:

1. Partially jacketed valve for low-temperature non-critical processes.
2. Fully jacketed valve with oversize flanges. This jacket provides uniform temperature control and is used mainly for high-temperature processes.
3. Fully jacketed valve with standard flanges. This is only used with components or piping having swaged jackets or special bolting facilities.
4. Fully jacketed valve having oversize flanges and the face-to-face dimension of an unjacketed valve. This construction should be used with caution. It is not recommended for new installations.

Advantages and disadvantages

Integrally-jacketed piping systems and components have long been the preferred method used with processes that require elevated temperatures for efficient in-plant transfer or product.

Advantages of integral jacketing include:

- unit construction
- high rates of heat transfer from the heating medium to the process
- ability to maintain processing temperatures within close tolerances

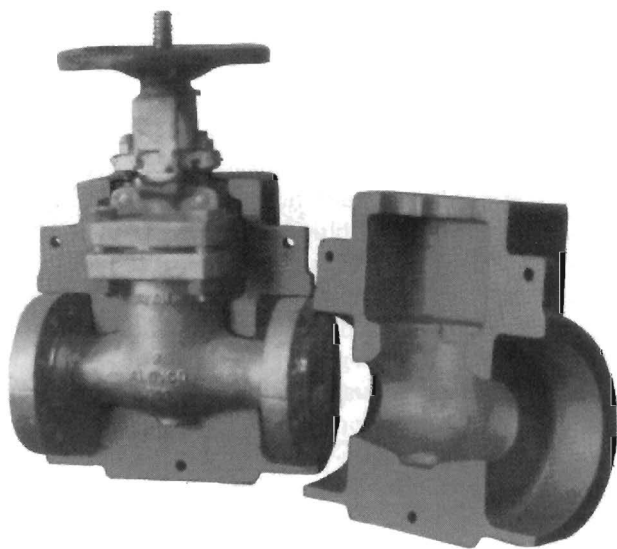


Figure 4. Typical two-piece ControHeat valve jacket.

Like many systems, they have their disadvantages; these include:

- the limited selections available for jacketed components
- relatively long deliveries for these components
- inconsistencies of quality of the jacketed components due to the lack of industry-wide fabrication standards

An alternative to integrally-jacketed piping systems is the clamp-on heating system comprising bolt-on jackets for valves and heat-tracing elements for piping.

Practically any piece of equipment can be heated with a clamp-on jacket. However, the cost of the clamp-on system can increase as the required temperature of the process approaches the temperature of the heating fluid.

Component programs using finite element modelling have been produced by jacketed-piping fabricators to assist end-users in determining the right system or product for their application. Typically, these programs consider the thermal conductivities of the system components, film coefficients of both the processes and the heating fluid, a detailed temperature profile of the piping system, and the heat lost to the atmosphere through the insulation and the net heat input to the process.

Dual-containment piping

Dual- or double-containment piping is by no means a new concept to process piping applications. Systems using a carrier pipe with secondary containment have been installed using metal pipes for many years. Applications have included systems for the nuclear and chemical process industries where hazardous or highly toxic materials have been transported.

Dual-containment piping is all about employee safety and environmental damage caused by chemical spills and leaks. It is widely accepted that pollution of the soil due to chemical leaks eventually leads to groundwater contamination which has become a major world-wide problem.

One proven method of avoiding serious leaks from piping conveying hazardous or highly toxic materials is to utilise a primary pipe contained within a protective outer casing, combined with a leak-detection system to pinpoint leaks and simultaneously raise an alarm.

Systems of this type have often proven to be complex, labour intensive and costly.

Polypropylene and PVDF piping are now widely and successfully used for transporting chemical waste and toxic and other hazardous materials (Figure 5).

Typically, the design uses a twin-wall pipe which is extruded as a single homogeneous structure. The fittings are often moulded as a one-piece item, with both inner and outer walls fixed in place.

The pipes and fittings may be joined with standard butt welding with a single weld simultaneously joining both the inner and outer walls.